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JEE Main Physics Practice Sheet — DPP-052

Topic: Angular Momentum — Definition, Cross Product & Relation with Torque

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A particle of mass $m = 2$ kg is located at $\vec{r} = (3\hat{i} + 4\hat{j})$ m and moves with a linear momentum $\vec{p} = (2\hat{i} - 3\hat{j})$ kg · m/s. The angular momentum $\vec{L} = \vec{r} \times \vec{p}$ of the particle about the origin is:

(A) $-17\hat{k}$ kg · m^2/s

(B) $+17\hat{k}$ kg · m^2/s

(C) $-7\hat{k}$ kg · m^2/s

(D) $+7\hat{k}$ kg · m^2/s

Q2. The fundamental dynamical equation connecting external torque $\vec{\tau}$ and angular momentum $\vec{L}$ of a system of particles about an inertial point is:

(A) $\vec{\tau} = \vec{L} \times \vec{v}$

(B) $\vec{L} = \dfrac{d\vec{\tau}}{dt}$

(C) $\vec{\tau} = \dfrac{d\vec{L}}{dt}$

(D) $\vec{\tau} = \vec{r} \times \vec{L}$

Q3. A particle of mass m is projected with an initial velocity u at an angle θ with the horizontal from the origin $(0, 0)$. The magnitude of the angular momentum of the projectile about the launch point at any time t during flight is:

(A) $mugt^2 \cos\theta$

(B) $mugt \sin\theta$

(C) $\dfrac{1}{2}mgut^2 \cos\theta$

(D) $\dfrac{1}{2}mgut^2 \sin\theta$

Q4. A particle moves along the line $y = d$ parallel to the x-axis with a constant velocity $\vec{v} = v\hat{i}$. The magnitude of its angular momentum about the origin:

(A) Increases linearly with time

(B) Decreases inversely with time

(C) Remains constant and equal to mvd

(D) Depends on the x-coordinate of the particle

Q5. The angular momentum of a system of particles about a point is conserved if and only if:

(A) The total kinetic energy of the system is constant

(B) The net external torque acting on the system about that point is zero

(C) The net external force acting on the system is zero

(D) The system is executing uniform circular motion

Q6. The angular momentum vector $\vec{L} = \vec{r} \times \vec{p}$ of a particle is given by $\vec{L}(t) = (4t^2\hat{i} - 2t\hat{j} + 6\hat{k})$ J·s. The net torque acting on the particle at $t = 2$ s about the origin is:

(A) $(8\hat{i} - 2\hat{j})$ N · m

(B) $(16\hat{i} + 2\hat{j})$ N · m

(C) $(16\hat{i} - 2\hat{j} + 6\hat{k})$ N · m

(D) $(16\hat{i} - 2\hat{j})$ N · m

Q7. A particle of mass m is executing uniform circular motion in the xy-plane of radius R with constant angular velocity $\vec{\omega} = \omega\hat{k}$. The angular momentum of the particle about the center of the circle is:

(A) $mR^2\omega\hat{k}$

(B) $-mR^2\omega\hat{k}$

(C) $\dfrac{1}{2}mR^2\omega\hat{k}$

(D) $\vec{0}$

Q8. The physical dimensions of angular momentum are identical to the dimensions of:

(A) Linear momentum

(B) Torque

(C) Planck's constant (h)

(D) Power

Q9. A particle of mass $m = 1$ kg is dropped from rest from point $(d, 0, h)$ at $t = 0$. The magnitude of the angular momentum of the particle about the origin $(0, 0, 0)$ as a function of time t is:

(A) $mgdt^2$

(B) $\dfrac{1}{2}mgdt$

(C) $mgdt$

(D) $mght$

Q10. A planet of mass m revolves around the Sun in an elliptical orbit. As the planet moves from aphelion (farthest distance r_a, speed v_a) to perihelion (nearest distance r_p, speed v_p), the relation between distances and speeds is:

(A) $r_a v_p = r_p v_a$

(B) $r_a^2 v_a = r_p^2 v_p$

(C) $r_a v_a^2 = r_p v_p^2$

(D) $r_a v_a = r_p v_p$

Q11. A conical pendulum consists of a bob of mass m rotating in a horizontal circle of radius R with constant speed v. The magnitude of the angular momentum of the bob about the center of the circular path is:

(A) mvR

(B) $mv\sqrt{R^2 + h^2}$

(C) mvh

(D) Zero

Q12. If a central force $\vec{F} = f(r)\hat{r}$ acts on a particle, the torque on the particle about the center of force is:

(A) Maximum

(B) Zero, hence angular momentum about the center is strictly conserved

(C) Variable and depends on r

(D) Proportional to r^2

Q13. A particle of mass $m = 2$ kg moves with velocity $\vec{v} = (3\hat{i} + 4\hat{j})$ m/s along the straight line $4x - 3y = 12$. The magnitude of angular momentum about the origin is:

(A) $12 \text{ kg} \cdot \text{m}^2/\text{s}$

(B) $18 \text{ kg} \cdot \text{m}^2/\text{s}$

(C) $48 \text{ kg} \cdot \text{m}^2/\text{s}$

(D) $24 \text{ kg} \cdot \text{m}^2/\text{s}$

Q14. Kepler's second law of planetary motion (areal velocity $\dfrac{dA}{dt}$ = constant) is a direct consequence of the conservation of:

(A) Linear momentum

(B) Kinetic energy

(C) Total mechanical energy

(D) Angular momentum

Q15. A particle of mass m is projected with speed u at an angle of 45° to the horizontal. The magnitude of the angular momentum of the particle about the point of projection when it hits the ground is:

(A) $\dfrac{mu^3}{\sqrt{2}g}$

(B) $\dfrac{mu^3}{2g}$

(C) $\dfrac{mu^3}{g}$

(D) $\dfrac{\sqrt{2}mu^3}{g}$

Q16. The position vector of a particle is $\vec{r} = a\cos(\omega t)\hat{i} + b\sin(\omega t)\hat{j}$. The torque $\vec{\tau}$ acting on the particle about the origin is:

(A) $\vec{0}$

(B) $-m\omega^2(a\hat{i} + b\hat{j})$

(C) $m\omega^2 ab\hat{k}$

(D) $-m\omega^2 ab\hat{k}$

Q17. A disc of mass M and radius R is rotating about its fixed central axis with angular velocity $\vec{\omega}$. The angular momentum vector $\vec{L}$ of the disc:

(A) Lies in the plane of the disc

(B) Is parallel to the angular velocity vector $\vec{\omega}$ along the axis of rotation

(C) Is perpendicular to the axis of rotation

(D) Precesses continuously in the horizontal plane

Q18. A particle moves in the xy-plane with position $\vec{r} = (2t\hat{i} + 3t^2\hat{j})$ m. If the mass of the particle is $m = 1$ kg, the angular momentum about the origin as a function of time t is:

(A) $+6t^2\hat{k} \text{ kg} \cdot \text{m}^2/\text{s}$

(B) $+12t^2\hat{k} \text{ kg} \cdot \text{m}^2/\text{s}$

(C) $+6t^3\hat{k} \text{ kg} \cdot \text{m}^2/\text{s}$

(D) $+12t\hat{k} \text{ kg} \cdot \text{m}^2/\text{s}$

Q19. If the net external force on a system is zero ($\sum \vec{F}_{\text{ext}} = \vec{0}$), but the net external torque is non-zero ($\sum \vec{\tau}_{\text{ext}} \neq \vec{0}$), then:

(A) Linear momentum is conserved, but angular momentum is not conserved

(B) Angular momentum is conserved, but linear momentum is not conserved

(C) Both linear and angular momentum are conserved

(D) Neither linear nor angular momentum is conserved

Q20. A light rigid rod of length L has two equal masses m attached at its ends. It is rotating in a horizontal plane with constant angular speed ω about a vertical axis passing through its midpoint. The total angular momentum of the system about the axis is:

(A) $mL^2\omega$

(B) $\dfrac{1}{2}mL^2\omega$

(C) $\dfrac{1}{4}mL^2\omega$

(D) $2mL^2\omega$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-052)

Angular Momentum — Definition, Cross Product & Relation with Torque

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	D	11	A	16	A
2	C	7	A	12	B	17	B
3	C	8	C	13	D	18	A
4	C	9	C	14	D	19	A
5	B	10	D	15	A	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (A)
The angular momentum vector:

$$\begin{aligned}\vec{L} &= \vec{r}\times\vec{p} = (3\hat{i}+4\hat{j})\times(2\hat{i}-3\hat{j})\\ &= (3(-3)-4(2))\hat{k} = (-9-8)\hat{k}\\ &= -17\hat{k}\ \text{kg}\cdot\text{m}^2/\text{s}\end{aligned}$$

Sol 2. Option (C)
Differentiating $\vec{L}=\vec{r}\times\vec{p}$ with respect to time:

$$\frac{d\vec{L}}{dt} = \left(\frac{d\vec{r}}{dt}\times\vec{p}\right)+\left(\vec{r}\times\frac{d\vec{p}}{dt}\right)$$

Since $\frac{d\vec{r}}{dt}\times\vec{p} = \vec{v}\times(m\vec{v}) = \vec{0}$:

$$\frac{d\vec{L}}{dt} = \vec{r}\times\vec{F}_{\text{ext}} = \vec{\tau}_{\text{ext}}$$

Sol 3. Option (C)
Position vector of the projectile at time t:

$$\vec{r}(t) = (u\cos\theta\, t)\hat{i} + \left(u\sin\theta\, t - \frac{1}{2}gt^2\right)\hat{j}$$

Velocity vector:

$$\vec{v}(t) = (u\cos\theta)\hat{i} + (u\sin\theta - gt)\hat{j}$$

Angular momentum $\vec{L} = m(\vec{r}\times\vec{v})$:

$$\begin{aligned}\vec{L} &= m(xv_y - yv_x)\hat{k}\\ &= m\left[(u\cos\theta t)(-gt) + \frac{1}{2}gt^2(u\cos\theta)\right]\hat{k}\\ &= -\left(\frac{1}{2}mgut^2\cos\theta\right)\hat{k}\end{aligned}$$

Magnitude is $\frac{1}{2}mgut^2\cos\theta$.

Sol 4. Option (C)
Position vector $\vec{r} = x\hat{i} + d\hat{j}$ and momentum $\vec{p} = mv\hat{i}$.

$$\vec{L} = (x\hat{i}+d\hat{j})\times(mv\hat{i}) = -mvd\hat{k}$$

The perpendicular distance from origin to line of motion is d, so magnitude $|\vec{L}| = mvd =$ constant.

Sol 5. Option (B)
From $\vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}$, if $\vec{\tau}_{\text{ext}} = \vec{0}$, then $\frac{d\vec{L}}{dt} = \vec{0} \implies \vec{L} =$ constant.

Sol 6. Option (D)
Given $\vec{L}(t) = 4t^2\hat{i} - 2t\hat{j} + 6\hat{k}$:

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \frac{d}{dt}(4t^2\hat{i}-2t\hat{j}+6\hat{k}) = 8t\hat{i} - 2\hat{j}$$

At $t = 2$ s:

$$\vec{\tau}(2) = 8(2)\hat{i} - 2\hat{j} = (16\hat{i} - 2\hat{j})\ \text{N}\cdot\text{m}$$

Sol 7. Option (A)
For a particle in circular motion:

$$\vec{L} = \vec{r}\times\vec{p} = \vec{r}\times(m\vec{v})$$

Since $\vec{v} = \vec{\omega}\times\vec{r} \implies v = \omega R$ perpendicular to $\vec{r}$:

$$L = mvR = m(\omega R)R = mR^2\omega\hat{k}$$

Sol 8. Option (C)
Dimensions of angular momentum:

$$[L] = [r][p] = [\text{L}][\text{M L T}^{-1}] = [\text{M L}^2\text{T}^{-1}]$$

Planck's constant $h = \frac{E}{\nu} \implies [h] = \frac{[\text{M L}^2\text{T}^{-2}]}{[\text{T}^{-1}]} = [\text{M L}^2\text{T}^{-1}]$.

Sol 9. Option (C)
After falling for time t, downward speed is $v = gt$.
Linear momentum is $p = mgt$ vertically downward along $x = d$.
Perpendicular distance to the line of motion is $r_\perp = d$.
Magnitude of angular momentum:

$$L = p\cdot r_\perp = (mgt)d = mgdt$$

Sol 10. Option (D)
The gravitational force between the Sun and the planet is purely central ($\vec{\tau} = \vec{0}$), conserving angular momentum:

$$L = mr_av_a = mr_pv_p \implies r_av_a = r_pv_p$$

Sol 11. Option (A)
About the center of the circular orbit, radius is R and velocity v is purely tangential ($v\perp R$).

$$L = mvR$$

Sol 12. Option (B)
For any central force, $\vec{F}\parallel\vec{r}$, so:

$$\vec{\tau} = \vec{r}\times\vec{F} = \vec{r}\times(f(r)\hat{r}) = \vec{0}$$

Since $\vec{\tau} = \vec{0}$, angular momentum about the center of force is conserved.

Sol 13. Option (D)
Perpendicular distance from origin to line $4x - 3y - 12 = 0$:

$$r_\perp = \frac{|-12|}{\sqrt{4^2 + (-3)^2}} = \frac{12}{5} = 2.4 \text{ m}$$

Speed is $v = \sqrt{3^2 + 4^2} = 5$ m/s.
Angular momentum magnitude:

$$L = mvr_\perp = (2 \text{ kg})(5 \text{ m/s})(2.4 \text{ m}) = 24 \text{ kg} \cdot \text{m}^2/\text{s}$$

Sol 14. Option (D)
Areal velocity is $\frac{dA}{dt} = \frac{L}{2m}$. Since the gravitational force is central, $\vec{\tau} = \vec{0} \implies L =$ constant, keeping areal velocity constant.

Sol 15. Option (A)
Range is $R = \frac{u^2 \sin(2\times 45^\circ)}{g} = \frac{u^2}{g}$.
At landing, downward vertical velocity component is $v_y = u \sin 45^\circ = \frac{u}{\sqrt{2}}$.
Perpendicular lever arm for vertical momentum about origin is the horizontal range R:

$$L = p_y \cdot R = (mv_y)R = m\left(\frac{u}{\sqrt{2}}\right)\left(\frac{u^2}{g}\right) = \frac{mu^3}{\sqrt{2}g}$$

Sol 16. Option (A)
Acceleration vector:

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} = -\omega^2(a\cos\omega t\hat{i} + b\sin\omega t\hat{j}) = -\omega^2\vec{r}$$

The force is central: $\vec{F} = m\vec{a} = -m\omega^2\vec{r}$.
Torque about origin:

$$\vec{\tau} = \vec{r} \times \vec{F} = \vec{r} \times (-m\omega^2\vec{r}) = \vec{0}$$

Sol 17. Option (B)
For a symmetrical body rotating about a principal axis of symmetry, $\vec{L} = I\vec{\omega}$, which is strictly collinear with the angular velocity vector along the axis of rotation.

Sol 18. Option (A)
Velocity vector: $\vec{v} = \frac{d\vec{r}}{dt} = 2\hat{i} + 6t\hat{j}$.
Angular momentum with $m = 1$ kg:

$$\begin{aligned}\vec{L} &= m(\vec{r} \times \vec{v}) = (2t\hat{i} + 3t^2\hat{j}) \times (2\hat{i} + 6t\hat{j}) \\ &= (2t(6t) - 3t^2(2))\hat{k} = (12t^2 - 6t^2)\hat{k} = +6t^2\hat{k} \text{ kg} \cdot \text{m}^2/\text{s}\end{aligned}$$

Sol 19. Option (A)
Linear momentum changes according to $\sum \vec{F}_{\text{ext}} = \frac{d\vec{P}}{dt} = \vec{0} \implies \vec{P} =$ constant.
Angular momentum changes according to $\sum \vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt} \neq \vec{0} \implies \vec{L} \neq$ constant.

Sol 20. Option (B)
Each mass is at distance $r = L/2$ from the axis.
Moment of inertia: $I = m(L/2)^2 + m(L/2)^2 = 2m(L^2/4) = \frac{1}{2}mL^2$.
Total angular momentum:

$$L = I\omega = \frac{1}{2}mL^2\omega$$